

# Samsung Innovation Campus

Choose **any real-world dataset or data source** and build a small cloud-based analytics solution using **Microsoft Fabric**.

Your solution must include the following:

## Requirements

### 1. Create a Fabric Workspace

- Create a dedicated workspace for your project.

#### Create a workspace

Name \*

Fabric\_Student\_Project

✓ This name is available

Description

datasets containing student scores, study hours, and academic metrics into a structured OneLake Lakehouse. The data is transformed into Delta tables to feed Power BI dashboards for educational tracking and insights

Domain ⓘ

Assign to a domain (optional)

[Learn more about workspace settings](#) ↗

Workspace image



↑ Upload

↶ Reset

Advanced ^

Contact list \* ⓘ

Amr.Morsi.4234043 (Owner) ×

Workspace type ⓘ

Apply

Cancel

Activate Windows

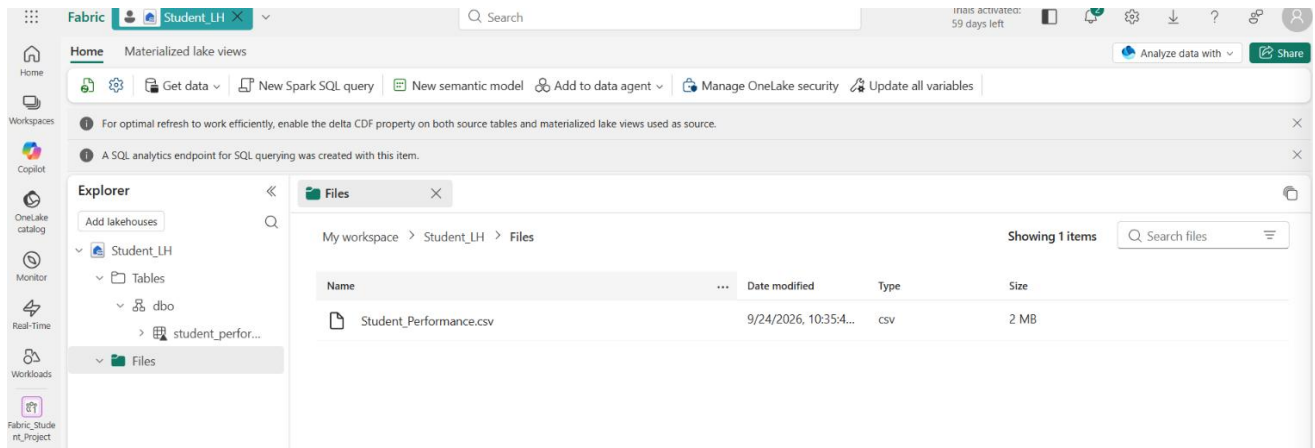
Go to Settings to activate Windows.

### 2. Bring Data into Fabric

- Use at least **one external data source**. (Student\_Performance\_2.csv)
- The source can be a CSV/Excel file, database, API, SaaS application, or any other suitable source.
- Use a Fabric ingestion method to load the data.

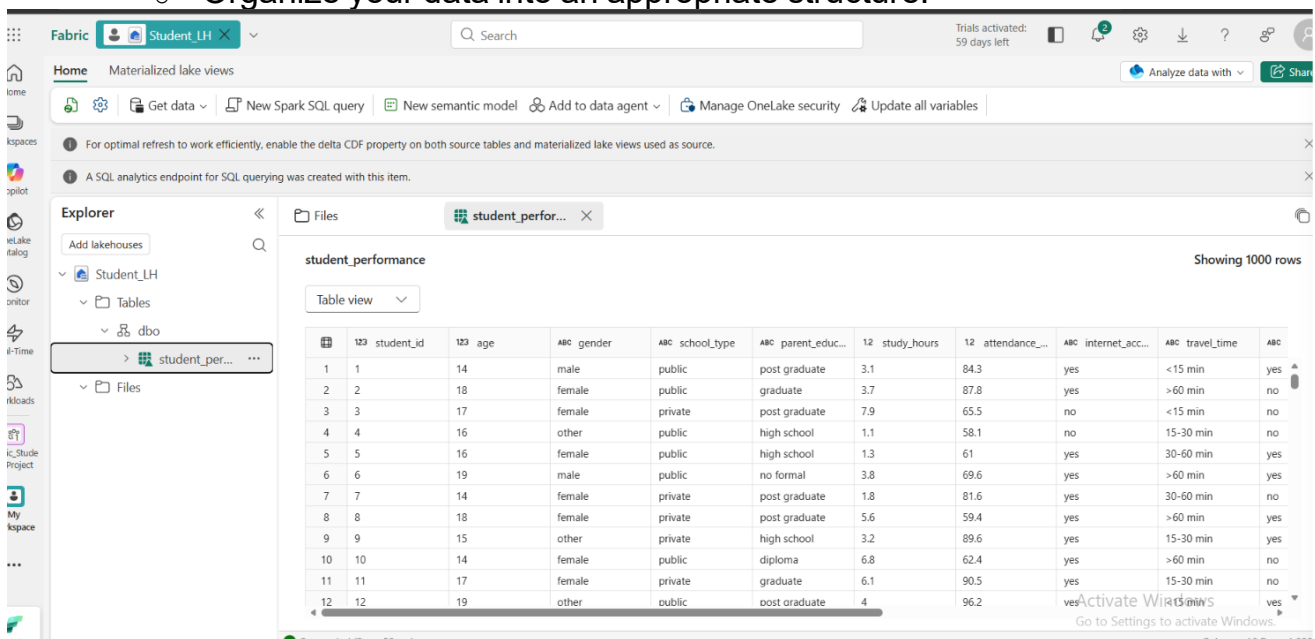
# Samsung Innovation Campus

Ya Handsa I Converted the raw file into an optimized Delta Lake table using the native "Load to Tables" ingestion feature



## 3. Store the Data in OneLake

- Store your data inside Fabric.
- The project must use **OneLake-backed storage**.
- Organize your data into an appropriate structure.



## 4. Process the Data

- Perform data cleaning and transformation using **at least one Fabric processing tool**.
- You can use SQL, PySpark, Dataflow Gen2, or another suitable Fabric capability.

# Samsung Innovation Campus

- Filtered and cleaned the student data to remove missing or invalid values.
- 

Files | student\_perform... | Query 1

Run | Save as view | Open in notebook

```
1 SELECT
2     student_id,
3     gender,
4     study_hours,
5     attendance_percentage,
6     extra_activities,
7     math_score,
8     science_score,
9     english_score,
10    overall_score,
11    final_grade
12 FROM dbo.student_performance
13 WHERE overall_score IS NOT NULL
14    AND study_hours >= 0;
```

Table view | Download | Filter by keyword

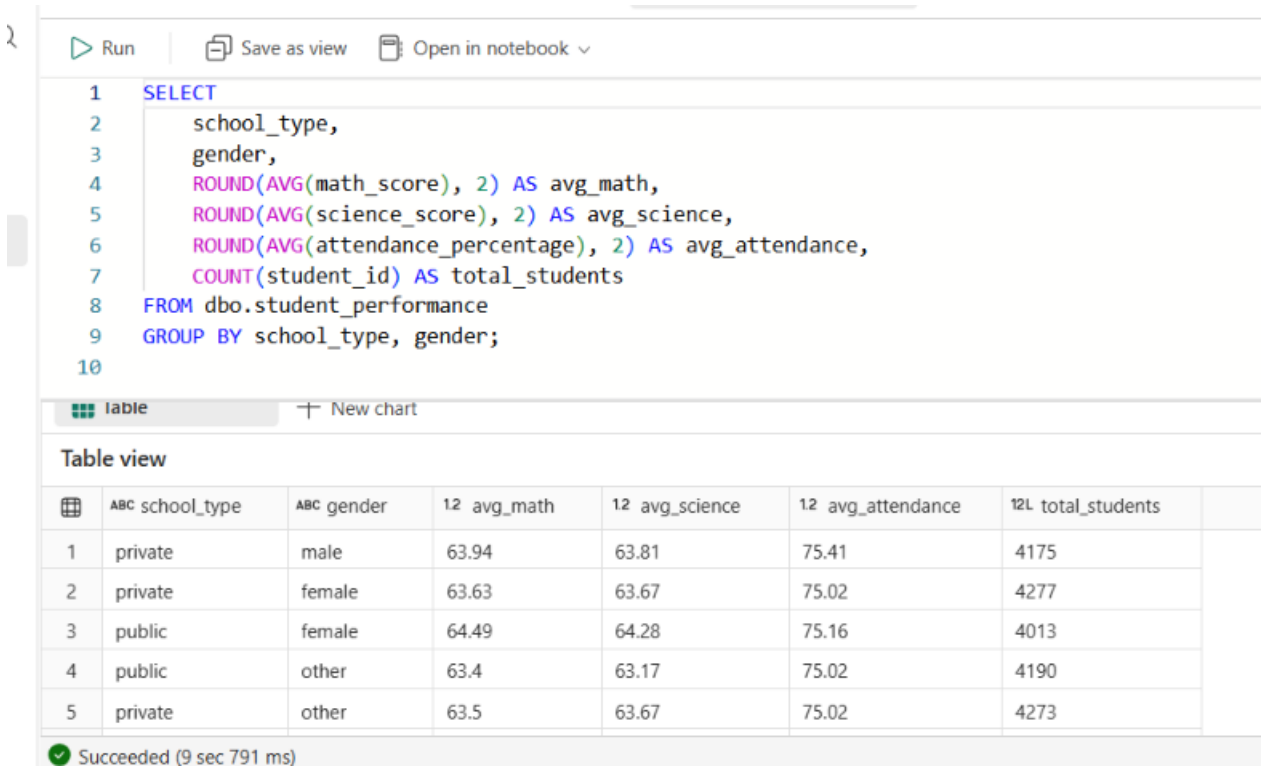
|    | 123 student_id | ABC gender | 12 study_hours | 12 attendance_percentage | ABC extra_activities | 12 math_score | 12 science_score | 12 english_score | 12 overall_score |
|----|----------------|------------|----------------|--------------------------|----------------------|---------------|------------------|------------------|------------------|
| 1  | 1              | male       | 3.1            | 84.3                     | yes                  | 42.7          | 55.4             | 57               | 53.1             |
| 2  | 2              | female     | 3.7            | 87.8                     | no                   | 57.6          | 68.8             | 64.8             | 61.3             |
| 3  | 3              | female     | 7.9            | 65.5                     | no                   | 84.8          | 95               | 79.2             | 89.6             |
| 4  | 4              | other      | 1.1            | 58.1                     | no                   | 44.4          | 27.5             | 54.7             | 41.6             |
| 5  | 5              | female     | 1.3            | 61                       | yes                  | 8.9           | 32.7             | 30               | 25.4             |
| 6  | 6              | male       | 3.8            | 69.6                     | yes                  | 51.5          | 78.3             | 63.9             | 63.5             |
| 7  | 7              | female     | 1.8            | 81.6                     | no                   | 41.9          | 29.4             | 39.2             | 39.1             |
| 8  | 8              | female     | 5.6            | 59.4                     | yes                  | 56.7          | 60.1             | 53.4             | 69.6             |
| 9  | 9              | other      | 3.2            | 89.6                     | yes                  | 54.1          | 59.5             | 38.3             | 55.2             |
| 10 | 10             | female     | 6.8            | 62.4                     | no                   | 71.9          | 70.4             | 81.3             | 69.6             |
| 11 | 11             | female     | 6.1            | 90.5                     | no                   | 75.4          | 82.7             | 70.8             | 77.5             |

Inspect

# Samsung Innovation Campus

Aggregated average math and science scores alongside attendance rates grouped by school type and gender.

To highlight performance variations across student groups to guide targeted academic support



Aggregated subject-level scores across predefined student attendance tiers

And the objective of this query to Quantified the direct relationship between classroom attendance consistency and student academic success.

# Samsung Innovation Campus

The **Objective of this** Provided a standardized benchmark to evaluate overall student achievement for downstream reporting.

Then I used the py spark to run this query

RunSave as viewOpen in notebook

```
1 SELECT
2     extra_activities,
3     ROUND(AVG(study_hours), 2) AS avg_study_hours,
4     ROUND(AVG(overall_score), 2) AS avg_overall_score,
5     COUNT(student_id) AS total_students
6 FROM dbo.student_performance
7 GROUP BY extra_activities;
```

TableNew chart

Table view

|   | ABC extra_activities | 12 avg_study_hours | 12 avg_overall_score | 12L total_students |
|---|----------------------|--------------------|----------------------|--------------------|
| 1 | no                   | 4.22               | 63.86                | 12500              |
| 2 | yes                  | 4.28               | 64.15                | 12500              |

```
1 %%pyspark
2 from pyspark.sql import functions as F
3
4 df = spark.read.table("dbo.student_performance")
5 cleaned_df = df.dropDuplicates(["student_id"])
6 transformed_df = cleaned_df.withColumn(
7     "avg_core_score",
8     F.round((F.col("math_score") + F.col("science_score") + F.col("english_score")) / 3.0, 2)
9 )
10 display(transformed_df.select("student_id", "school_type", "study_hours", "avg_core_score", "final_grade"))
```

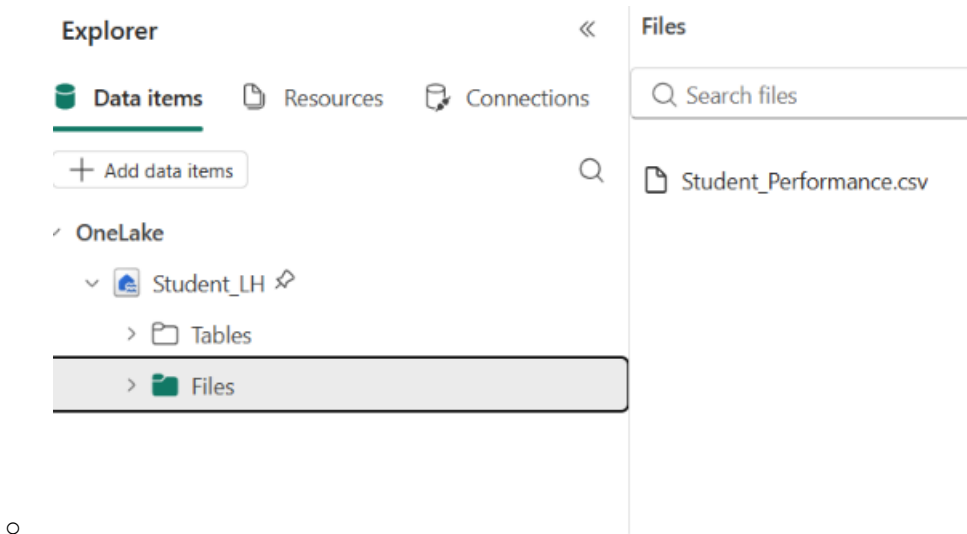
✓ 28 sec - Command executed in 26 sec 89 ms by Amr Abdel-Monem Sanoussi Morsi on 9/25/2026, 12:33:42 AMPySpark (Python)

## 5. Use at Least Two Data Representations

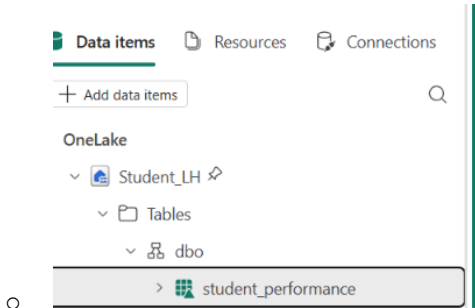
Your project must contain at least **two of the following**:

- Files

# Samsung Innovation Campus



- Lakehouse Tables



| Table view |                |                 |                |                   |              |
|------------|----------------|-----------------|----------------|-------------------|--------------|
|            | 123 student_id | ABC school_type | 12 study_hours | 12 avg_core_score | ABC final_gr |
| 1          | 1              | public          | 3.1            | 51.7              | e            |
| 2          | 2              | public          | 3.7            | 63.73             | d            |
| 3          | 3              | private         | 7.9            | 86.33             | b            |
| 4          | 4              | public          | 1.1            | 42.2              | e            |
| 5          | 5              | public          | 1.3            | 23.87             | f            |
| 6          | 6              | public          | 3.8            | 64.57             | d            |
| 7          | 7              | private         | 1.8            | 36.83             | f            |
| 8          | 8              | private         | 5.6            | 56.73             | d            |

- Warehouse Tables

- Views

Explorer

Run

Save as view

Open in notebook

1

CREATE OR REPLACE VIEW dbo.v\_high\_performers AS

2

SELECT student\_id, overall\_score, final\_grade

3

FROM dbo.student\_performance

4

WHERE overall\_score >= 80;

5

SELECT \* FROM dbo.v\_high\_performers LIMIT 10;

6

Table

New chart

Table view

|   | 123 student_id | 12 overall_score | ABC final_grade |  |
|---|----------------|------------------|-----------------|--|
| 1 | 3              | 89.6             | b               |  |
| 2 | 21             | 81.3             | c               |  |
| 3 | 24             | 94.6             | b               |  |
| 4 | 26             | 92.5             | b               |  |
| 5 | 27             | 82.5             | c               |  |
| 6 | 30             | 82.6             | c               |  |

## 6. Create an Analytical Model

- Prepare data specifically for analysis.
- These photos for fact and dim tabels

```
2 CREATE OR REPLACE VIEW dbo.dim_students AS
3 SELECT DISTINCT
4     student_id,
5     gender,
6     age,
7     school_type,
8     parent_education,
9     internet_access
10 FROM dbo.student_performance;
11 CREATE OR REPLACE VIEW dbo.dim_study_habits AS
12 SELECT DISTINCT
13     student_id,
14     study_hours,
15     attendance_percentage,
16     extra_activities,
17     study_method
18 FROM dbo.student_performance;
19 CREATE OR REPLACE VIEW dbo.fact_student_performance AS
20 SELECT
21     student_id,
22     math_score,
23     science_score,
24     english_score,
25     overall_score,
26     final_grade
27 FROM dbo.student_performance;
```

- You may use a star schema, dimensional model, or another appropriate design.

This photo after create fact and dimensional tables

# Samsung Innovation Campus

```
1 SELECT
2     d.school_type,
3     d.gender,
4     h.extra_activities,
5     ROUND(AVG(f.overall_score), 2) AS avg_score,
6     COUNT(f.student_id) AS total_students
7 FROM dbo.fact_performance f
8 JOIN dbo.dim_student d ON f.student_id = d.student_id
9 JOIN dbo.dim_habits h ON f.student_id = h.student_id
10 GROUP BY d.school_type, d.gender, h.extra_activities;
```

 Table  New chart

Table view

|  | ABC school_type | ABC gender | ABC extra_activities | 12L avg_score | 12L total_students |
|-----------------------------------------------------------------------------------|-----------------|------------|----------------------|---------------|--------------------|
| 1                                                                                 | private         | female     | no                   | 63.45         | 2155               |
| 2                                                                                 | public          | other      | no                   | 63.63         | 2068               |
| 3                                                                                 | public          | male       | yes                  | 63.96         | 2068               |
| 4                                                                                 | private         | other      | no                   | 63.84         | 2140               |

## 7. Build a Power BI Report

- Create a Power BI report connected to your Fabric data.

Direct Lake semantic model name ⓘ

sm\_student\_analytics

Workspace ⓘ

 Fabric\_Student\_Project ▼

Select or deselect tables for the semantic model.

 Search  

☐ Select all

▼  dbo

☒  dim\_habits

☒  dim\_student

☒  fact\_performance

☐  student\_performance

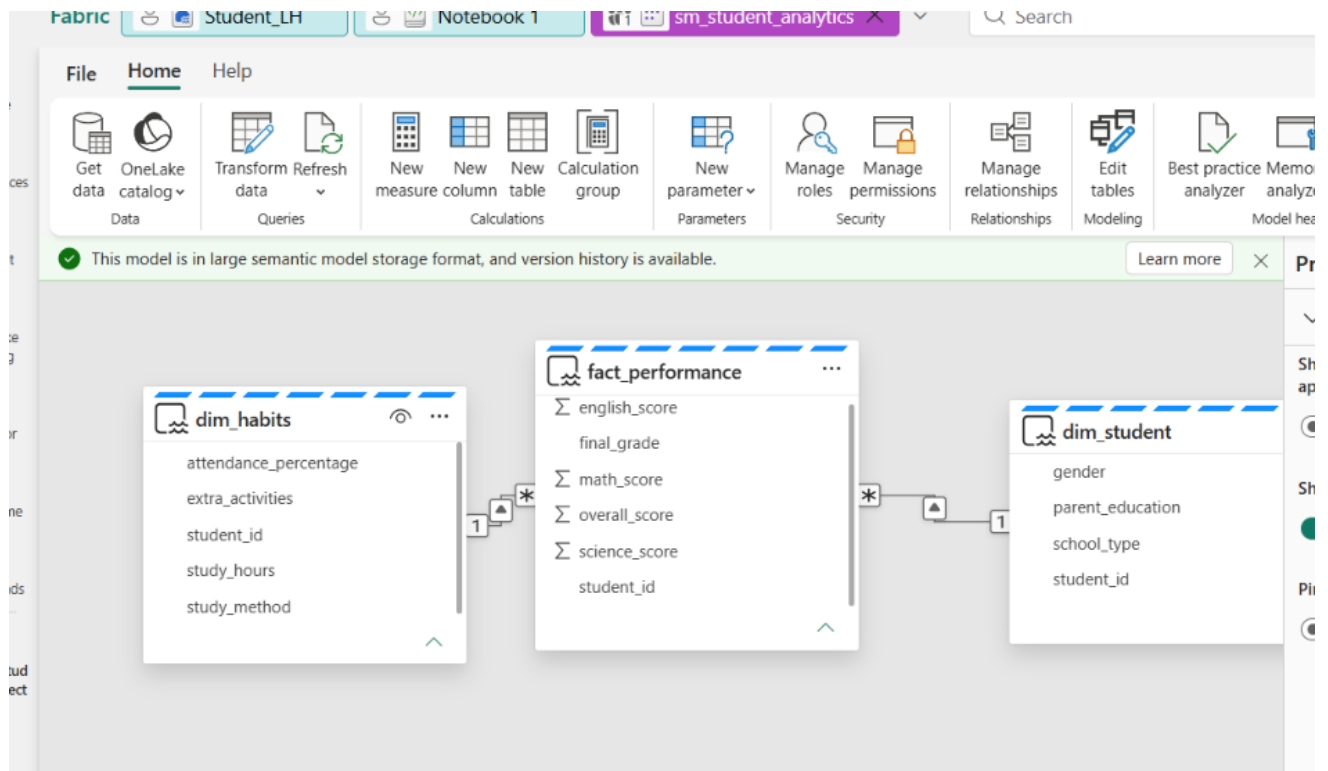
[Learn more about Direct Lake](#) 

○ Please wait...

Confirm

# Samsung Innovation Campus

## Star schema



- Include at least **3 meaningful analytical questions/KPIs**.

**Card KPI (Average of overall\_score):** Displays the baseline academic performance across all students with an average score of 64.01.

**Top-Right Chart (Math & Science Scores by Gender):** Compares aggregate math and science performance across genders, showing nearly identical achievement levels.

**Bottom-Left Chart (Overall Score by Extra Activities):** Evaluates overall score distribution between students participating in extracurricular activities and those who do not.

**Bottom-Right Chart (Overall Score by School Type):** Benchmarks cumulative student achievement, showing higher total overall scores in private schools compared to public schools.

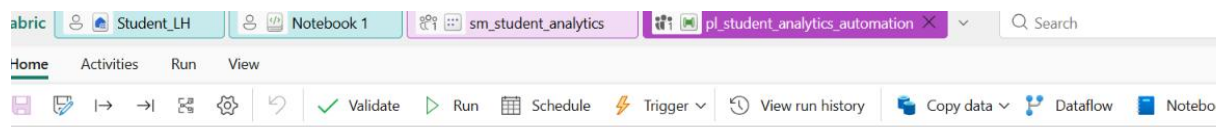
# Samsung Innovation Campus



○

## 8. Automate the Solution

- Create a **Pipeline** that runs part of your solution.



### Build a pipeline to organize and move your data

#### Start with a blank canvas



##### Pipeline activity

Automate data orchestrations using rich no-code activities.

#### Start with guidance



##### Copy data assistant

Follow guided steps to copy data into Microsoft Fabric, as well as other data stores.



##### Practice with sample data

Quickly build a pipeline with a predefined template to load data into Lakehouse.



##### Templates

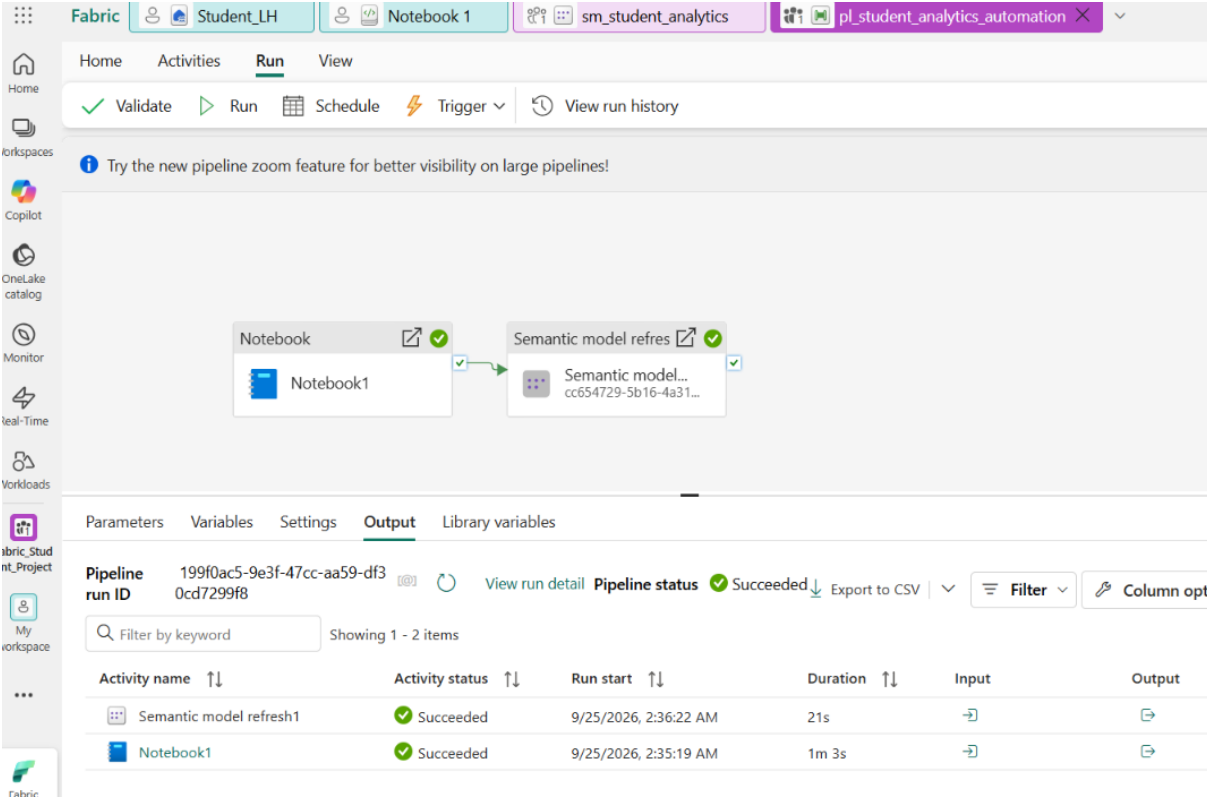
Generate a new pipeline quickly using a predefined data scenario.

Need help? Ask Copilot

○

- The pipeline should contain at least **two activities**.
- Configure a trigger or schedule so the pipeline can run automatically.

# Samsung Innovation Campus



The screenshot displays the Microsoft Fabric interface. At the top, there are tabs for 'Fabric', 'Student\_LH', 'Notebook 1', 'sm\_student\_analytics', and 'pl\_student\_analytics\_automation'. The 'Run' tab is active, showing a pipeline diagram with two activities: 'Notebook1' and 'Semantic model refresh1'. Below the diagram, the 'Output' tab is selected, showing a table with the following data:

| Activity name           | Activity status | Run start             | Duration | Input | Output |
|-------------------------|-----------------|-----------------------|----------|-------|--------|
| Semantic model refresh1 | Succeeded       | 9/25/2026, 2:36:22 AM | 21s      | →     | →      |
| Notebook1               | Succeeded       | 9/25/2026, 2:35:19 AM | 1m 3s    | →     | →      |

## 9. Demonstrate Cloud Connectivity

Your project must use at least **one additional connection/integration**, such as:

- Another database
- API

# Samsung Innovation Campus

```
1 %%pyspark
2 import requests
3 import json
4 url = "https://jsonplaceholder.typicode.com/users"
5 response = requests.get(url, timeout=10)
6 data = response.json()
7
8 rdd = spark.sparkContext.parallelize([json.dumps(user) for user in data])
9 api_df = spark.read.json(rdd)
10
11 api_df.select("id", "name", "email", "phone", "website") \
12     .write.format("delta").mode("overwrite").saveAsTable("dbo.external_cloud_users")
13
14
15 display(spark.sql("SELECT * FROM dbo.external_cloud_users LIMIT 5"))
```

✓ 26 sec - Command executed in 26 sec 210 ms by Amr Abdel-Monem Sanoussi Morsi on 9/25/2026, 2:47:44 AM

> Spark jobs (14 of 14 succeeded) Resources Log

Table

New chart

Table view Download

|   | 12L id | ABC name      | ABC email     | ABC phone     | ABC website   |  |
|---|--------|---------------|---------------|---------------|---------------|--|
| 1 | 9      | Glenna Rei... | Chaim_Mc...   | (775)976-6... | conrad.com    |  |
| 2 | 10     | Clementina... | Rey.Padber... | 024-648-3...  | ambrose.net   |  |
| 3 | 2      | Ervin Howell  | Shanna@m...   | 010-692-6...  | anastasia.net |  |
| 4 | 3      | Clementine... | Nathan@y...   | 1-463-123-... | ramiro.info   |  |
| 5 | 7      | Kyle M...     | Tyler M...    | 248-867-6123  | l...          |  |

Act  
Go t

- 
- SaaS application
- External storage
- Another dataset/source

External Cloud Dataset (GitHub Open **Storage Integration**).

# Samsung Innovation Campus

```
16 import pandas as pd
17
18 github_url = "https://raw.githubusercontent.com/datasets/gdp/master/data/gdp.csv"
19 pdf = pd.read_csv(github_url)
20
21 external_df = spark.createDataFrame(pdf.head(100))
22 external_df.write.format("delta").mode("overwrite").saveAsTable("dbo.external_github_gdp")
23
24 display(spark.sql("SELECT * FROM dbo.external_github_gdp LIMIT 5"))
```

11 sec - Command executed in 11 sec 295 ms by Amr Abdel-Monem Sanoussi Morsi on 9/25/2026, 2:50:10

> Spark jobs (14 of 14 succeeded) Resources Log

Table New chart

Table view

|   | 12L id | ABC name        | ABC email      | ABC phone     | ABC website  |  |
|---|--------|-----------------|----------------|---------------|--------------|--|
| 1 | 9      | Glenna Rei...   | Chaim_Mc...    | (775)976-6... | conrad.com   |  |
| 2 | 10     | Clementina...   | Rey.Padber...  | 024-648-3...  | ambrose.net  |  |
| 3 | 4      | Patricia Leb... | Julianne.O...  | 493-170-9...  | kale.biz     |  |
| 4 | 5      | Chelsey Di...   | Lucio_Hetti... | (254)954-1... | demarco.info |  |

## 10. Security

- Configure appropriate workspace/item permissions.

I get my friend the permission to be a viewer

sm\_student\_analytics

Related content

- Dashboards
- Reports
- Workbooks

+ Add user

Links Direct access Access via related items

| People and groups with access   | Email Address                           | Permissions                           |
|---------------------------------|-----------------------------------------|---------------------------------------|
| Amr Abdel-Monem Sanoussi Morsi  | Amr.Morsi.4234043@HCBA.hnu.edu.eg       | Workspace Admin, All permissions g... |
| Yaseen Mahmoud Mustafa Mohammed | Yaseen.Mohammed.4234070@HCBA.hnu.edu.eg | Workspace Viewer                      |

- If applicable to your dataset, implement **Row-Level Security** in the analytical/reporting layer.

•

## 11. Monitoring

# Samsung Innovation Campus

- Run your pipeline and demonstrate how you can check whether the execution succeeded or failed.

home

rispaces

topilot

retake

atalog

onitor

al-Time

rkloads

ric\_Stude

Project

My

rispace

...

Monitoring

Monitor

Activities

Configure notifications

Schedule failures

Preview

Monitor

View and track the status of the activities across all the workspaces for which you have permissions within Microsoft Fabric.

Refresh

Filter by keyword

Filter

Column Options

Clear all

To apply filters, select the values from the Filter dropdown menu.

| Activity name                                  | Status    | Item type      | Start time           | Submitted by          | Location           |
|------------------------------------------------|-----------|----------------|----------------------|-----------------------|--------------------|
| Notebook 1_961f34d6-ab72-4ce5-...              | Stopped   | Notebook       | 09/25/2026, 2:41 AM  | Amr Abdel-Monem Sanou | My workspace       |
| sm_student_analytics                           | Succeeded | Semantic model | 09/25/2026, 2:36 AM  | Amr Abdel-Monem Sanou | Fabric_Student_... |
| Notebook 1_e4888a4c-1da6-44fc-852e-c08bfa0...  | Succeeded | Notebook       | 09/25/2026, 2:35 AM  | Amr Abdel-Monem Sanou | My workspace       |
| pl_student_analytics_automation                | Succeeded | Pipeline       | 09/25/2026, 2:35 AM  | Amr Abdel-Monem Sanou | Fabric_Student_... |
| Notebook 1_0382727f-f246-479c-a4ac-2c876e5...  | Stopped   | Notebook       | 09/25/2026, 2:09 AM  | Amr Abdel-Monem Sanou | My workspace       |
| Notebook 1_61f045b8-218a-4b64-88a7-6e06f3a...  | Stopped   | Notebook       | 09/25/2026, 12:28 AM | Amr Abdel-Monem Sanou | My workspace       |
| HC_Student_LH_2f7c7497-6aa2-43c4-a38b-a3fbf... | Succeeded | Lakehouse      | 09/24/2026, 11:57 PM | Amr Abdel-Monem Sanou | My workspace       |
| HC_Student_LH_LakehouseOperation_36ed6c79-...  | Succeeded | Lakehouse      | 09/24/2026, 11:28 PM | Amr Abdel-Monem Sanou | My workspace       |

Showing all available data

Activate Windows  
Go to Settings to activate Windows.

## 12. Data Lineage

# Samsung Innovation Campus

- Demonstrate the lineage of your data from the source to the final analytical/reporting layer.

